

Google Ads API — Design documentation

Tool name

nthdesigns Prospect Scanner — CPC Benchmark Module

Company

Company: nthdesigns

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1. Purpose

The nthdesigns Prospect Scanner is an internal business development tool used by nthdesigns staff to identify UK local businesses with weak digital presence (Google Business Profile and website accessibility), generate audit reports, and draft personalised outreach emails.

The Google Ads API integration is a small, optional module that fetches local keyword cost-per-click (CPC) benchmarks to support GBP-focused outreach copy. Example: "Businesses in your category in Birmingham spend approximately £X per click on Google Ads..."

The tool does not create, manage, or modify Google Ads campaigns, ad groups, ads, or keywords in Google Ads.

2. Users and access

- Audience: nthdesigns employees only (authenticated internal web app).
- Access model: Login required (Laravel Breeze). No public API. No third-party or client access.
- Deployment: Private server (Laravel Cloud). Not listed on Google Workspace Marketplace or any public directory.

3. Google Ads API usage

Permissible use requested: Researching keywords and recommendations

API version: v18 (REST)

Services and methods used (read-only)

Service	Method / endpoint	Purpose
KeywordPlanIdeaService	customers/{customerId}:generateKeywordIdeas	Fetch average CPC metrics for locally seeded commercial keywords
GeoTargetConstantService	geoTargetConstants:suggestGeoTargetConstants	Resolve city + country to a geo target constant for local CPC lookup

Services NOT used: CampaignService, AdGroupService, AdService, KeywordService (mutations), BillingService, or any mutate/write operations.

Typical request flow

1. Operator enters niche (e.g. "dental practice") and city (e.g. "Birmingham") in the internal UI.
2. App builds 3–5 seed keywords (e.g. "dental practice Birmingham", "dental practice in Birmingham").
3. App calls suggestGeoTargetConstants to resolve "Birmingham, United Kingdom" → geo target ID (cached in config when possible).
4. App calls generateKeywordIdeas with seed keywords, English language constant, and geo target.
5. App reads keywordIdeaMetrics.averageCpcMicros (fallback: top-of-page bid micros), computes median CPC, stores result in our database.
6. CPC is used only when generating internal outreach email drafts. It is not written back to Google Ads.

Trigger: Operator-initiated only (button: "Fetch from Google Ads"). Auto-fetch on search create is disabled by default (GOOGLE_ADS_CPC_AUTO_FETCH=false).

4. Data handling

Stored locally (PostgreSQL):

- Median CPC value (£)
- Source label (google_ads)
- Seed keywords used for the lookup
- Geo target resource name

Not stored: Full Google Ads API responses, competitor data, or personally identifiable information from Google Ads.

CPC data is used solely to enrich internal outreach email drafts. Operators may override or enter CPC manually without calling the API.

5. Authentication and security

- OAuth 2.0 with refresh token (scope: <https://www.googleapis.com/auth/adwords>).

Credentials stored in server environment variables only:

- GOOGLE_ADS_CLIENT_ID
- GOOGLE_ADS_CLIENT_SECRET
- GOOGLE_ADS_REFRESH_TOKEN
- GOOGLE_ADS_DEVELOPER_TOKEN
- GOOGLE_ADS_CUSTOMER_ID
- Access tokens cached server-side (~55 minutes). Refresh tokens are never exposed to the browser or end users.
- All API calls originate from the application server, not from client browsers.

6. Error handling and logging

- API failures are logged server-side (`google_ads.cpc_lookup_failed`, `google_ads.geo_target_failed`) with niche/city context — no credentials in logs.
- Failures do not block other app functionality; operators can enter CPC manually.
- No automatic retries beyond standard HTTP client behaviour.

7. Expected API volume

- Estimated usage: 5–50 keyword lookups per week during active outreach periods.
- Operations per lookup: 1–2 API calls (`suggestGeoTargetConstants` when geo not cached, plus `generateKeywordIdeas`).
- Daily estimate: Well under 100 operations/day — far below Basic Access limits (15,000/day).

8. What the tool does NOT do

- Does not create, edit, pause, or delete campaigns, ad groups, ads, or keywords in Google Ads.
- Does not access billing, conversion, or performance reporting beyond keyword idea metrics.
- Does not provide Google Ads management features to external users or clients.
- Does not resell or redistribute Google Ads data.

9. Architecture overview

Operator (browser)

- Laravel web app (authenticated)
- MarketCpcLookupService / GoogleAdsKeywordPlanService
- Google Ads REST API (read-only keyword planning)
- PostgreSQL (market_cpc_defaults, searches)
- Outreach email generator (uses stored CPC in AI prompt)

Stack: Laravel 13, PostgreSQL, internal Inertia/React UI. REST client (no third-party Google Ads PHP library).

10. Compliance

- Internal use only — tool used exclusively by nthdesigns for its own prospecting.
- RMF: Tool is exempt (under 15,000 requests/day, no external campaign management product).
- Operators send outreach manually outside the app; the tool generates drafts only.